

PAVAN KUMAR UNDAVALLI

Python Backend Developer

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Professional Summary

- Backend developer with 3.5+ years of experience in Python development, automation, and scalable backend systems.
- Expertise in building FastAPI-based REST services, AI application wrappers, and integrating LLMs and RAG pipelines.
- Experience developing full-stack AI systems using FastAPI, PostgreSQL/pgvector, LangChain, and Streamlit.
- Strong in scalable backend architecture, automation, and AI-driven platform development.

Technical Skills

Languages: Python, SQL

Backend: FastAPI, REST API Development, OOP

Generative AI: LangChain, OpenAI APIs, RAG, Prompt Engineering

Databases: PostgreSQL, pgvector, Vector Databases

Tools: Git, GitHub, Postman, VS Code

Frontend: Streamlit

Concepts: API Development, Automation, Data Structures, Scalable Backend Architecture

Experience

Wipro

July 2024 – Nov 2025

Senior Associate — Python Backend & GenAI Developer

Hyderabad, India

- Developed production-grade REST APIs using **FastAPI** to serve machine learning and Generative AI models.
- Built API wrapper layers over LLM and data science applications for unified secure access.
- Integrated **LangChain, OpenAI APIs, and RAG pipelines** into backend systems for conversational AI and document querying.
- Built AI solutions using **FastAPI, PostgreSQL/pgvector, and Streamlit**.
- Implemented vector database pipelines for embeddings storage and semantic search.
- Worked with data science teams to deploy and productionize ML/GenAI models into API services.
- Implemented logging, validation, and performance tuning for production APIs.

RMSI

Jan 2022 – Jan 2024

Python Developer

Hyderabad, India

- Developed Python automation scripts and backend utilities for workflow automation.
- Built REST APIs and backend services for internal tools and integrations.
- Worked on data processing pipelines and database operations.
- Improved performance and maintainability of backend systems.

Key Projects

AI Document Query System

2025

FastAPI + LangChain + PostgreSQL

- Built RAG-based document query system using LangChain and OpenAI APIs.
- Integrated pgvector for semantic search and embeddings storage.
- Developed FastAPI backend and Streamlit interface.

Automation & API Integration Tools

2024

Python + FastAPI

- Built automation scripts and backend APIs for integrations and workflow automation.
- Designed modular backend architecture for scalable deployment.

Education

Bachelor of Engineering (ECE)

2019

Sir C.R. Reddy College of Engineering, Eluru

Diploma in ECE

2014

Sri Varalakshmi Polytechnic College